

WORKED KEY

Full Name: _____

Period: _____

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. Write a numerical expression that represents six less than the difference of 24.5 and 7. DO NOT EVALUATE.

Answer: $(24.5 - 7) - 6$

2. Evaluate the expression from problem 1.

Answer: $17.5 - 6 = 11.5$ (accept a correct evaluation of the student's own #1)

3. The table shows the prices at the Cardinal concession stand. At Friday's game the Ruiz family bought 3 pretzels, 2 hot dogs, and 4 waters. Write and evaluate a numerical expression for the total cost.

Item	Price
Pretzel	\$2.50
Hot dog	\$3.75
Water	\$1.25

Answer: $3(2.50) + 2(3.75) + 4(1.25) = 7.50 + 7.50 + 5.00 = \20.00

4. Which expression represents each verbal expression? Write the letter. One choice is never used.

- (A) $5m - 2$
 (B) $2 - 5m$
 (C) $5(2 - m)$
 (D) $2 - (5 + m)$
 (E) $5m + 2$

a. the product of 5 and m, subtracted from 2 _____

Answer: B

b. five times the difference of 2 and m _____

Answer: C

c. two less than the product of 5 and m _____

Answer: A

d. 2 decreased by the sum of 5 and m _____

Answer: D

5. Evaluate $3(a + b)^2 \div 4 + 2c$, if $a = 1$, $b = 3$, $c = 7$.

Answer: $3(4)^2 \div 4 + 2(7) = 48 \div 4 + 14 = 12 + 14 = 26$

6. The table shows how much Priya earns working at summer camp. Write an algebraic expression for her earnings for leading a archery sessions and c canoe sessions. Then find her earnings if she leads 6 archery sessions and 3 canoe sessions.

Job	Pay
Leading an archery session	\$12.00 each
Leading a canoe session	\$15.00 each

Answer: $12a + 15c$; $12(6) + 15(3) = 72 + 45 = \117

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7. Name the property that makes each statement true.

a. $2 \cdot (7 \cdot y) = (2 \cdot 7) \cdot y$

Answer: Associative Property of Multiplication

b. $r \cdot 1 = r$

Answer: Identity Property of Multiplication

8. Simplify: $10c + 7 - 6c - 2$

Answer: $4c + 5$

9. Simplify: $6(2x + 5) - 4x$

Answer: $12x + 30 - 4x = 8x + 30$

10. Simplify: $9p - 5(p - 4)$

Answer: $9p - 5p + 20 = 4p + 20$

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11. Error Analysis — Rex used the Distributive Property on $-7(3x + 4)$ and wrote $-21x + 28$. Identify his mistake, then show the correct result.

Answer: the -7 makes BOTH products negative; correct: $-21x - 28$

12. A rectangular parking lot has width w yards, and its length is $4w - 2$ yards. Write a simplified expression for its perimeter. Then find the perimeter if $w = 3$.

Answer: $P = 2(4w - 2) + 2w = 10w - 4$; $w = 3 \rightarrow P = 26$ yd

13. Write an algebraic expression for the verbal expression: seven more than one-third of k .

Answer: $k/3 + 7$

Cumulative Review — ACT Style. Circle the best answer. No work required, but scratch work never hurts.

14. $5/8 + 1/6 = ?$

(A) $6/14$

(B) **$19/24$**

(C) $5/48$

(D) $3/4$

Answer: $15/24 + 4/24 = 19/24$

15. 60% of 45 is what number?

(E) 20

(F) 25

(G) **27**

(H) 30

Answer: $0.60(45) = 27$

16. $-6 - (-9) + 4 = ?$

(A) -11

(B) **7**

(C) -1

(D) 19

Answer: $-6 + 9 + 4 = 7$